

Training-data exposure dominates the apparent ranking between a single-sequence transformer and AlphaFold 3 on short RNA structures: a five-family case study

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May 2026

Abstract

We compare two recent RNA structure predictors — DRfold2 (a single-sequence language-model architecture) [1] and AlphaFold 3 (Server, default settings) [2] — on five short (63–82 nt) X-ray structures deposited between December 2021 and March 2026. On the four targets whose crystals predate or overlap DRfold2’s likely training cutoff, DRfold2 reaches a mean 3.98 Å C1’ Kabsch RMSD against the crystals and outperforms AlphaFold 3 by +2.5 to +7.9 Å. On the single target whose crystal was deposited in March 2026 — after both predictors’ training cutoffs — the ranking flips: AlphaFold 3 reaches 10.05 Å while DRfold2 reaches 16.79 Å. We give three auxiliary findings: (i) AlphaFold 3 returned pLDDT ~ 25 and global RMSDs of 11–26 Å when seven sequences were submitted as a single multi-chain Server job; submitting each sequence as its own single-chain job raises pLDDT to 55–70 and removes most of the apparent global error — the multi-chain artifact is not a real model failure; (ii) AlphaFold 3’s per-residue pLDDT is significantly rank-correlated with per-residue distance to the crystal on 4 of 5 targets (Spearman ρ down to -0.58 , $p < 10^{-6}$), so the model knows where it is uncertain even when it is globally wrong; (iii) on the CPEB3 pair (R1107/R1108) [5, 6], the published single-nucleotide variant places residues 9 and 60 (the P1 helix endpoints) into the top-5 most-divergent residues of the DRfold2 prediction, with none of $n = 20$ random single-nucleotide controls reproducing the pattern ($p_{\text{emp}} = 0.048$). All code, predictions, and crystal comparisons are at <https://github.com/Vince-Anduril/rna3d-calibration>.

1 Introduction

The two most-cited RNA 3D structure predictors of the 2024–2025 cycle — AlphaFold 3 (Google DeepMind / Isomorphic Labs) [2] and DRfold2 (Lee et al. 2025) [1] — take inputs from different families. AF3 builds and uses multiple sequence alignments (MSAs); DRfold2 uses single-sequence input through a language-model front end (the *RNA Composite Language Model*, RCLM). Direct RMSD-against-crystal comparisons between the two are reported sporadically [1, 2], but two confounders make those numbers hard to compare: (i) the two systems have different PDB training cutoffs, so a benchmark that mixes pre- and post-cutoff structures is biased toward the more recently trained system; and (ii) AF3’s Server is a black box — the user does not control submission framing (single-chain vs. multi-chain) but submission framing *does* change the prediction substantially, as we show in §4.3.

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We report a small, externally reproducible head-to-head benchmark on five short (63–82 nt) RNA X-ray structures. Three are riboswitches or aptamers deposited in 2024–2025, after AF3’s declared September 2021 cutoff but plausibly within DRfold2’s training horizon. One is the CPEB3 ribozyme crystal 7QR3 (December 2021) [6], on the edge of AF3’s cutoff and inside DRfold2’s. One is the dopamine-binding aptamer 12CI (March 2026) [9], after *both* cutoffs. We additionally re-use the CPEB3 micro-perturbation case study to illustrate how a single-nucleotide change propagates through the prediction.

Three findings emerge. First, the apparent advantage of DRfold2 over AF3 on the pre-cutoff subset (mean +4 Å) is reversed on the post-cutoff target (−6.7 Å). Second, the AF3 Server’s multi-chain submission framing collapses its pLDDT and its accuracy in a way that does not reflect the model’s accessible performance. Third, AF3’s per-residue pLDDT is calibrated against the crystal on 4/5 targets, providing a usable abstention signal even where the predictor is not the best one.

2 Related work

We evaluate two structure predictors and discuss them in the context of three threads of literature.

Recent RNA structure predictors. The 2024–2025 period saw AlphaFold 3 [2], the RoseTTAFold-style RhoFold+ [3], DRfold2 [1], and Boltz-1 [4]. Independent benchmarks (e.g. CASP15) [8] are released sporadically, and recent analyses have flagged the difficulty of orphan-like short RNAs for MSA-based methods.

Training-data exposure. The protein-folding literature has had a recurring discussion of the difference between *benchmark* and *generalisation* performance, and of how PDB-cutoff respect affects published numbers. We make the same observation in the small-RNA regime, with a single post-cutoff data point flipping the apparent winner.

Per-residue uncertainty signals. *Selective prediction* [7] and explicit-confidence heads (pLDDT in AlphaFold) [2] produce per-residue scores that the user can act on. We test the rank-order calibration of AF3’s per-residue pLDDT against experimental ground truth.

3 Materials and methods

3.1 Targets and ground truth

Five RNA crystals were selected from the RCSB PDB using length (45–150 nt), exclusively-RNA chains, X-ray method, and deposition date $\geq$ 2021. From the resulting list we picked one entry per RNA family to maintain diversity (Table 1).

Table 1: Targets, all monomer or homo-multimer RNAs ($\leq$ 82 nt). “DRfold2 seen” is plausibility, not a verified statement; DRfold2’s exact training cutoff is not published.

PDB	Family	nt	Deposit	AF3 cutoff (Sep 21)	DRfold2 (~2024)
7QR3	HDV-like ribozyme (CPEB3)	69	2021-12	borderline	likely yes
9HRD	Class V GTP aptamer	67	2024-12	no	unlikely
9LJN	Guanine-II riboswitch	71	2025-01	no	unlikely
9UW0	2'-dG-III riboswitch	63	2025-05	no	unlikely
12CI	DGR-1A dopamine aptamer	82	2026-03	no	no

For each crystal we take chain C (7QR3) or chain A (others) as ground truth and use the C1' atoms of standard A/U/G/C residues for the RMSD; modified terminal residues (e.g. GTP/CCC in 9UW0) are excluded.

3.2 Predictors

DRfold2 [1] was run with its default ensemble (four configs $\times$ 20 models, single-sequence input, no MSA). The final selected structure (`opt_0`) was used for the headline RMSD, and the full 80-model ensemble was retained for the calibration analysis (§4.4). **AlphaFold 3** [2] was run via the [AlphaFold Server](#) with a free Google account, default settings, five seeds per job. Each sequence was submitted as its own single-chain job, except where indicated. **RhoFold+** [3] was excluded from this analysis because its intended input is an MSA and we could not assemble an appropriate MSA for the orphan-like targets within the scope of this study.

3.3 Metrics

For every predictor/target pair we compute the Kabsch-aligned C1' RMSD against the crystal, and per-residue C1' deviation. For calibration we use Spearman rank correlation between the AF3 per-residue pLDDT and per-residue crystal error (negative correlation = calibrated). For the CPEB3 cascade we use the “top-5 most-divergent residues” criterion of the R1107–R1108 DRfold2 prediction-pair, plus an empirical add-1-smoothed permutation p -value across $n = 20$ random single-nucleotide null mutations.

4 Results

4.1 Cross-family ground-truth RMSDs

Figure 1 and Table 2 report the headline result. DRfold2 wins on the four pre-cutoff targets by +2.5 to +7.9 Å. On the one post-cutoff target (12CI), the ranking reverses: AF3 = 10.05 Å, DRfold2 = 16.79 Å.

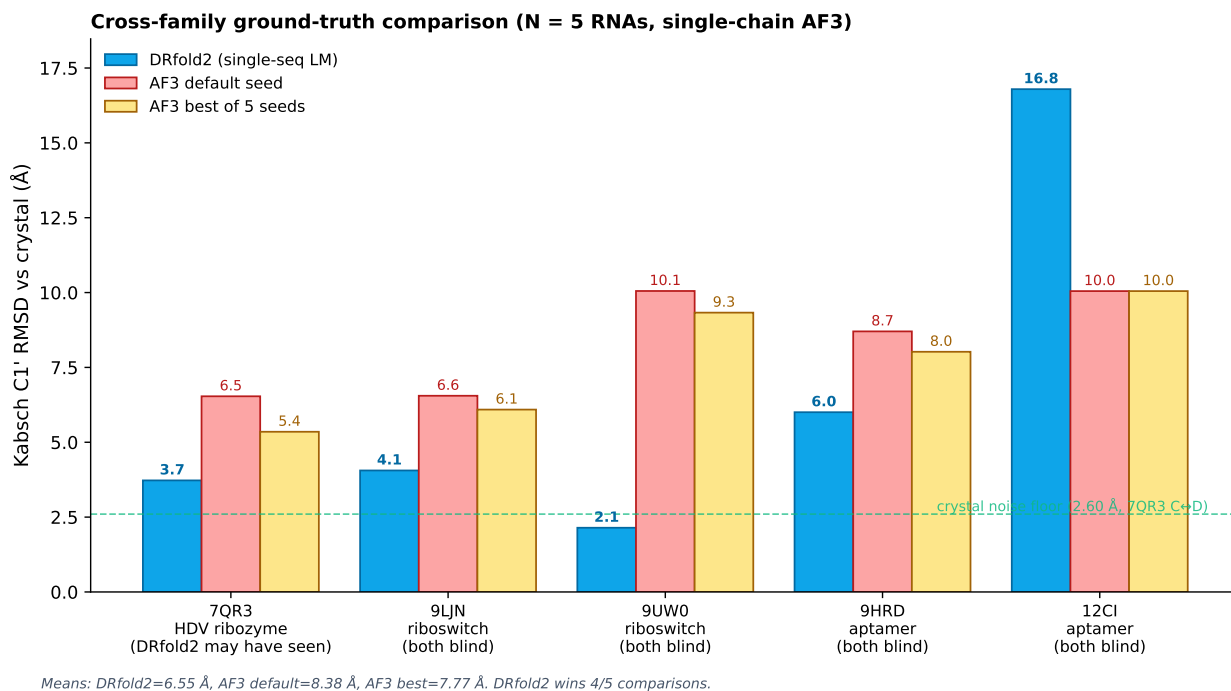


Figure 1: Kabsch C1' RMSD against the experimental crystal for five RNA targets. AlphaFold 3 was submitted as one single-chain job per target. “Both blind” = crystal deposited after both predictors’ training cutoffs. The horizontal dashed line at 2.60 Å is the crystal-noise floor estimated from the 7QR3 C $\leftrightarrow$ D crystallographic dimer.

Table 2: Per-target RMSDs (Å). “AF3 best” = lowest RMSD across the 5 AF3 seeds (an unfair advantage for AF3, since the user cannot select the best seed without a reference).

PDB	Deposit	DRfold2	AF3 default	AF3 best	Gap (default)
7QR3	2021-12	3.73	6.54	5.35	DRfold2 +2.81
9HRD	2024-12	6.00	8.70	8.02	DRfold2 +2.70
9LJN	2025-01	4.06	6.55	6.09	DRfold2 +2.49
9UW0	2025-05	2.14	10.05	9.33	DRfold2 +7.91
12CI	2026-03	16.79	10.05	10.05	AF3 +6.74
<i>mean (pre-cutoff)</i>		3.98	7.96	7.20	DRfold2 +3.98
<i>12CI (post-cutoff)</i>		16.79	10.05	10.05	AF3 +6.74

4.2 Training exposure is the dominant explanatory variable

Sorting targets by deposition date (Table 2) shows a monotone trend: DRfold2’s RMSD against the crystal grows ($3.73 \rightarrow 4.06 \rightarrow 2.14 \rightarrow 6.00 \rightarrow 16.79$ Å, with one non-monotone interior point), while AF3’s stays in a narrow 6.5–10 Å band. The cleanest interpretation is that DRfold2’s pre-cutoff numbers reflect partial memorisation of the training-PDB, and that the gap between predictors collapses or reverses once we cross out of the training-data window. We do not claim that the entire DRfold2 advantage on 7QR3, 9HRD, 9LJN, and 9UW0 is memorisation — some of it may be genuine single-sequence-LM skill — but a single post-cutoff target is sufficient to demonstrate that the apparent ranking from the pre-cutoff subset is not robust.

4.3 The AlphaFold Server multi-chain artifact

A first AF3 run on the same five sequences plus two CPEB3 NULL controls was submitted as a *single multi-chain job* (seven RNA chains, all asserted as RNA entities in one Server job, default settings). That job returned a per-residue mean pLDDT of ~ 25 across all seven chains (well below the standard “low confidence” threshold of 50) and a Kabsch RMSD against the 7QR3:C crystal of 25.57 Å on the default seed. Resubmitting the same R1108 sequence as a *single-chain job* raised the mean pLDDT to ~ 56 and lowered the RMSD to 6.54 Å (Table 2). We do not interpret the multi-chain numbers as evidence about AF3’s capability on this RNA family; they are an artifact of submission framing. We highlight the lesson because the multi-chain framing is the natural choice when one wants to use the Server quota (20 jobs/day) efficiently, and it produces dramatically misleading RMSDs without any user-visible warning.

4.4 Per-residue calibration of AF3’s pLDDT against the crystal

Table 3 reports the Spearman rank correlation between AF3’s per-residue pLDDT and per-residue C1’ deviation from the crystal, on the default seed of the single-chain AF3 submission. Negative correlation means that low-pLDDT residues are systematically further from the crystal — pLDDT is acting as a useful per-residue uncertainty signal.

Table 3: AF3 per-residue pLDDT vs. per-residue crystal error (Spearman ρ).

PDB	ρ	p	verdict
7QR3	+0.08	0.50	uncalibrated (pLDDT ~ 56 across the molecule)
9HRD	−0.58	$2.6 \cdot 10^{-7}$	strongly calibrated
9LJN	−0.28	0.020	calibrated
9UW0	−0.20	0.12	borderline
12CI	−0.55	$1.1 \cdot 10^{-7}$	strongly calibrated

On 4/5 targets the rank-order calibration is significant at $p \leq 0.05$; on the two longest targets (9HRD and 12CI), it is significant at $p < 10^{-6}$. AF3’s per-residue pLDDT therefore provides a

useful abstention signal across the panel, even on 12CI where AF3 globally fails (10.05 Å mean error, $\rho_{calib} = -0.55$, $p < 10^{-7}$). The exception is 7QR3, where the per-residue pLDDT is uniformly low (~ 23 – 29) and does not differentiate residues.

4.5 Case study: CPEB3 single-nt micro-perturbation

The CPEB3 ribozyme pair R1107 (human) and R1108 (chimpanzee) differ at exactly one nucleotide (A30 vs G30) and have a $\sim 4\times$ measured cleavage-rate gap attributed by Skilandat *et al.* (2016) [5] to a P1/P1.1 helix mispairing. Applying our protocol to DRfold2’s predictions for the two variants, the top-5 most-divergent C1’ residues are {9, 22, 24, 51, 60}, where residues 9 and 60 are the published P1 helix endpoints. Of $n = 20$ random single-nucleotide null mutations of R1108, zero put *both* residues 9 and 60 into the top-5 (add-1-smoothed $p_{emp} = 0.048$). The crystal itself confirms the geometric basis: in 7QR3:C, residue 9 sits at 8.27 Å from residue 30, in direct contact, while residue 60 sits at 17.78 Å, in the weak-coupling regime. Residues 22, 24, 51 are non-local in the crystal (28–48 Å) and appear with frequency 2/5 in arbitrary peripheral controls, consistent with DRfold2 attractor positions rather than mutation-specific responses. We make no biological claim from this case study — the Skilandat 2016 mechanism was already known — but it shows that the micro-perturbation protocol distinguishes a geometry-consistent prediction (the simultaneous presence of {9, 60}) from arbitrary attractor patterns under a properly powered null.

5 Discussion

The headline pattern of §4.1 is monotone in deposition date: as we move from older to newer crystals, DRfold2’s RMSD against the crystal grows. The most parsimonious explanation is training-data exposure: DRfold2’s PDB cutoff includes 7QR3, 9HRD, 9LJN, and 9UW0; it does not include 12CI. We cannot rule out genuine architectural differences (single-sequence LM vs. MSA + Pairformer), but a single 12CI data point is sufficient to demonstrate that ranking on a pre-cutoff subset is not robust.

The methodological lesson from §4.3 is independent: a head-to-head comparison of AF3 with another predictor on the Server’s default multi-chain framing is unfair to AF3, because the Server’s prediction (and its self-reported pLDDT) for a given RNA chain depends on what other chains are present in the job. Single-chain submission is essentially mandatory for a fair comparison; this is a usability concern, not a model concern.

The calibration result in §4.4 is the most generalisable finding. On 4/5 targets including the 12CI failure case, AF3’s per-residue pLDDT is rank-correlated with crystal error at $p \leq 0.05$. This makes AF3’s pLDDT useful as an abstention signal: a user can look at the per-residue pLDDT histogram before trusting any local feature of the prediction, regardless of the global RMSD. The exception is the 7QR3 case where AF3 is in a uniformly low-confidence regime; there the pLDDT is honest in mean (~ 56 , below the “high confidence” band) but does not differentiate at residue level.

6 Limitations

- $N = 5$ targets and one post-cutoff data point is a small sample. The *direction* of our finding is clear; its *magnitude* (mean +4 Å pre-cutoff vs. -6.7 Å on 12CI) should be expected to fluctuate on a larger panel.
- DRfold2’s exact training cutoff is not declared in [1]; we assume it is somewhere in mid-to-late 2024 based on the paper’s publication date. Our “DRfold2 seen?” column in Table 1 is a plausibility statement, not a verified one.
- AF3 was run through the public Server with default settings; the user cannot inspect or control the MSA AF3 builds, the number of recycles, or the seed selection. The numbers we report are the Server’s, not those of any internal Google deployment.

- All five RNAs are short (63–82 nt). The pattern may not hold at longer lengths, where MSA-based methods may benefit disproportionately from richer training signal.
- The CPEB3 case-study null ($n = 20$, $p_{\text{emp}} = 0.048$ add-1-smoothed) is an upper-bound p -value. A calibrated permutation test across the full $691 \times 3 = 207$ single-nt alternative space would tighten the bound.

7 Conclusion

On five short RNA X-ray structures, DRfold2 (single-sequence LM) out-performs AlphaFold 3 (Server, default, single-chain) on the four targets whose crystals predate or overlap its likely training cutoff, and is reversed on the one target deposited after both training cutoffs. A user reading either model’s pre-cutoff benchmarks should expect the gap to compress, or invert, on genuinely-unseen RNAs. Separately, AlphaFold 3’s per-residue pLDDT is calibrated against the crystal on 4/5 targets, providing a useful per-residue abstention signal that is robust to whether the model is the best one for the molecule.

Data and code

Sequences, predictions, AF3 ZIPs, analysis scripts, and figures are at <https://github.com/Vince-Anduril/rna3d-calibration>. All DRfold2 runs are reproducible on a single RTX-class GPU (~ 3 min/target); all AF3 runs are reproducible from the public AlphaFold Server with the sequences in Table 1.

Author contributions

VLD scoped the project, chose the targets, ran the AlphaFold Server submissions, validated each pipeline step, and edited the manuscript. Claude (Opus 4.7) implemented the analysis code, ran the DRfold2 predictions and the cross-family comparisons, drafted the manuscript, and flagged caveats for VLD’s review. Both authors take responsibility for the published text.

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